

Question	Answer	Marks
2(a)	- Weight is pointing downwards from centre of rod, - tension at end B is pointing upwards and along the axis of spring, - one force from the point A is pointing towards East-North direction, whose angle with the vertical is between 5 degrees to 35 degrees, - All 3 forces intersect at a common point. Deduct one mark if any of the above is wrong.	[1]
2(b)	Using the principle of moments and choosing A as the pivot, Anticlockwise moment of tension = clockwise moment of weight of rod $T \cdot 0.50 = W \cos 30^\circ \cdot 0.25$ $T \cdot 0.50 = 1.5 \cdot 9.81 \cdot \cos 30^\circ \cdot 0.25$ $T = 6.37$ $= 6.4 \text{ N}$	[1] [1]
2(c)	Resultant force is zero hence horizontal component of force at A is equal to horizontal component of tension. Horizontal component of force at A = $6.4 \sin 30^\circ = 3.2 \text{ N}$ to the left Resultant force is zero hence Vertical component of force at A + vertical component of tension = weight Vertical component of force at A = $1.5 \cdot 9.81 - 6.4 \cos 30^\circ = 9.17 \text{ N}$ upwards Magnitude of force = $\sqrt{3.2^2 + 9.17^2} = 9.71 \text{ N}$ $\tan \angle = 9.17/3.2 \rightarrow \theta = 71^\circ$ Direction: 139° anticlockwise from rod AB	[1] [1] [1]
2(d)	The point 'Z' is vertically <u>above the end B</u> of the rod but <u>longer than the original length BX</u>. Magnitude = half of weight of rod = 7.4 N	[1] [1]
	total marks	10

2020

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

Question	Answer	Marks
3(a)	Evidence of use of area below the line	[1]